...E.. R. Mulloy, continued from page 1.

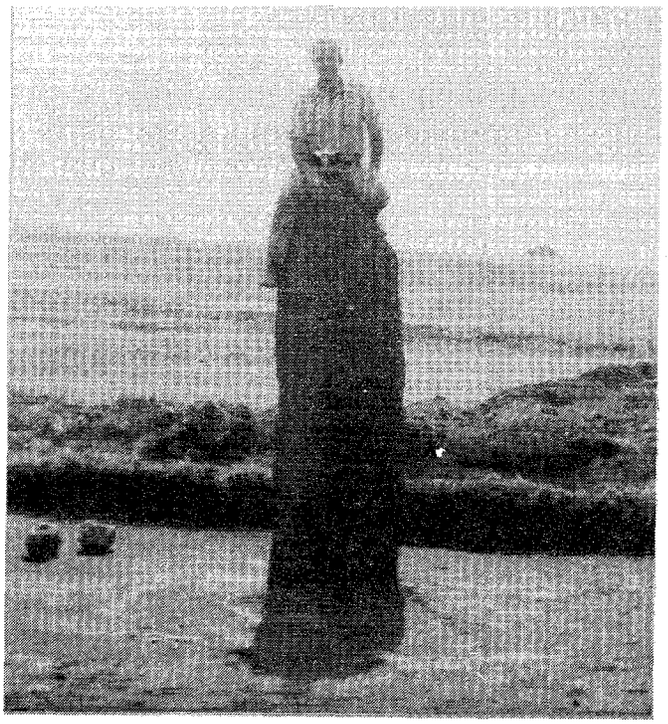
World" for which Bill acted as a consultant). Best of all was the surprise at a ceremony in the school grounds when he was presented with a beautiful hand-lettered and decorated document proclaiming him as an "Illustrious Citizen of Easter Island."

Of all the honors Bill received in his lifetime, this was the one he prized most highly. From his first visit in 1955 he had always had a strong affinity for the island and its people.

Bill had made more than 20 trips to the island over the years between 1955 and 1978, spending a total of nearly five years there; he had been planning, before his illness was diagnosed, to take early retirement and move to the island permanently that same year (1978). During his last season of field work on the island, a year and a half earlier, Bill had flown back to Santiago briefly to attend a seminar at the University of Chile's Institute of International Studies. The paper which he gave on this occasion, and which proved to be the last thing he ever wrote for publication, is entitled "A Preliminary Culture-Historical Research Model for Easter Island."

This paper, which is exerpted below, outlines a number of areas in which, he believed, more work was needed. Some of these have been studied in the past decade but as Bill frequently said, much more research is needed. The titles of many of his publications contain the words "preliminary" or "speculative", reflecting his caution about too hasty conclusions.

In a letter to Bill written 30 years ago by Father Sebastian Englert, who had just received reports of the carbon 14 dates from the excavations done by the 1955-56 Norwegian expedition, I found the following remarks which could equally well express Bill's own philosophy of science: "The Carbon 14 dates upsets all my chronological guesses in my book. But I should not worry about, because human science is a permanent correcting of errors."



William Mulloy relaxing atop discovery at Vinapu

A Preliminary Culture-Historical Research Model for Easter Island

William Mulloy

The paper from which the following excerpts have been taken has never be published in the U.S., but it did appear in 1979 in Vol.1 of Las Isl Oceanicas de Chile, *Instituto de Estudios Internacionales de la Unive sidad de Chile (pp. 105-151).*

SINCE ITS DISCOVERY on Easter Sunday of 1722 by the Dut navigator Jacob Roggeveen, the peculiar cultural characteristics Easter Island have fired the imaginations of almost all who have ha the opportunity to visit this tiny mote of land lost in the endle empty seas of the southeast Pacific. Much has been written about both by visitors and those who knew it only from the accounts others. From Roggeveen's day to this the compulsion to explain th unlikely presence of a spectacular and complex culture under su extreme conditions of isolation has been obsessive. Thousands titles exist in the literature ranging from accounts by casual visito of a few days or by those who had not seen the island at all to th works of serious scholars who systematically collected as many soli facts as they could and brought to bear on the problem the mo reasonable theoretical interpretations that could be derived fro these.

This disproportionate intensity of interest in such a tiny part of th world is readily understandable. It is one of the most basic canons history that complex cultures and civilizations, when certain oth conditions are met, are the product of relatively large populatio living in localities where cultural contacts with many other peopl are easy. Cultural complexity almost everywhere appears to b closely related to the intensity of opportunities to exchange ide with neighbors and to formulate new ones with the aid of these. Co versely, when such opportunities are lacking, one normally expec the simplest of cultural adaptations. The apparent violation of th basic principle on Easter Island has quite reasonably attracted th interest of many scholars and casual visitors as well.

It is probably not reasonable to characterize the culture of East Island as a civilization, though it has often been so describe Nevertheless, a culture with a written language (an undecipher hieroglyphic script that cannot be demonstrated to be related to an other in the world and thus, in the present state of knowledge at lea must be presumed to be a local development) possessed one of th most characteristic hallmarks of civilization. A class organized soc ety with enough coercive power to bring large work crews togeth and maintain them in systematic operation long enough to produ the remarkable amount of religious architecture and sculpture pre ent on the island certainly possessed a considerable degree political organization. The presence of solar ranging devices and th practice of orienting the facades of some structure to the azimuths the rising and setting sun at the solstices and equinox indica awareness of solar movements. These and other traits suggest a lev of cultural advancement amply sufficient, under such unusual co ditions of isolation, to have attracted the unusual attention which h been directed toward the island.

An understanding of these developments and other peculiariti is of considerable theoretical importance and clearly requires mu knowledge of the origin of the culture, of its pattern of developme on the island, of the relationship of its people to its tiny environme of 413 square kilometers and other kinds of information as we Unfortunately, unequivocal data are difficult to come by. Interpr tations of presently known facts suggest that the culture underwe

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an extended period of development toward greater complexity and eventually reached a climax which was followed by a period of population decrease, disorganization and decadence. All of this happened before the time of the first European contact. Thus, unlike the situation on many other Pacific islands where the classic form of the local culture was still a going concern at the time of the arrival of the first Europeans and where much information bearing on its non-material characteristics and the significance of material items could be recorded by eye-witnesses or recovered from informants who clearly remembered the events of a recent past, it appears that on Easter Island the events which terminated the period of climax were violent enough nearly to erase from the minds of the relatively few survivors much of the detail of the classic form of the culture and prevent its survival in legendary form.

Drastic reductions of population in the 19th Century due to devastating slave raids and a smallpox epidemic further reduced population and memories of the past before many systematic observers were present. In 1877 the total population of the island is said to be only 111 people. Such a small number could have retained only a fragment of the former cultural inventory and apparently most of the collective culture-historical memory was lost. Thus only a demoralized and to a considerable extent deculturated remnant population has been available as culture-historical informants and a disproportionately much greater reliance has had to be placed on archaeological evidence for an understanding of what went on here than has been true on many other Pacific islands. This lack of opportunity for direct observation of the classic form of the local culture has given peculiar emphasis to the emerging picture of the past of Easter Island.

The presently available body of archaeological evidence is far from adequate to permit a reliable and detailed general reconstruction of the local prehistory. Information about a "Settlement Period," embodying the earliest phase of the local development, is as yet entirely lacking. Though we have some information about domestic architecture, a coherent knowledge of it in a time perspective is lacking. Land-use patterns have been studied, though we have little understanding of the course of their development. Information about modifications of the environment due to human exploitation is sketchy though of central importance to an understanding of developments. Studies of artifacts through time and suitable for comparison with those of other islands are little developed. Best known are the large statues and the religious architecture. Here a temporal sequence in some detail is beginning to emerge. As a whole it is reasonable to say that, while presently available archaeological information is scanty and spotty, the potential for acquiring much more with continued investigation is good.

Because of the above circumstances it is impossible to provide a thoroughly reliable and detailed culture-historical summary. It is, however, possible to outline a sort of tentative model, not in the sense of a foreign historical sequence which local developments might be expected to parallel, but in the sense of a scenario which fits the scanty and incomplete local collection of information presently commonly taken to be fact. Such a model can be no more than one possible historical appreciation of the facts and would be expected continuously to be modified as new information became available. With such modifications it would be expected to approach progressively more closely an historical reality. Its principle value, beyond being a preliminary ordering of the materials, is as a research tool. It can serve to emphasize those areas where additional information is most badly needed and to suggest avenues and orientations for future work...

At this point Dr. Mulloy briefly discusses Métraux's and Englert's models , among others, and their "excessive attention to the problem of origin as a primary force in establishing the characteristics of the distinctive local developments..."

Written in the present state of knowledge, the above essay makes no pretense of conveying any precise or ultimate historical appreciation. It represents one reasonable interpretation of the scanty and incomplete collection of facts now available. Its principle value is as a research tool suggesting a variety of hypotheses that may be formulated and tested.

Among these is the notion that the environment has been progressively depleted by man since human occupation. Some evidence for this has already been accumulated but more detailed research by palaeobotanists is required. The extremely important studies of the pollen samples collected by the Heyerdahl expedition and made by Selling have not been published and perhaps should be repeated and extended.

Better and more detailed information is needed leading to an understanding of population development and the extent to which the carrying capacity of the island was approached. Central to this purpose is the completion of the ongoing archaeological survey especially with view to determining the actual number of domestic establishments present and their chronological provenience. Related to this is a determination of the intensity and probable productivity of former cultivation, especially in the interior of the island and the extent to which it was pushed into marginal areas. The archaeological survey will provide considerable additional information to this point from the recording of surface indications while paleobotanical and soil studies should also be useful. The analysis of a part of the survey evidence by Patrick McCoy represents a most useful approach and must be extended to other parts of the island. Studies of the amount and kind of potential marine resources such as those begun by...the Cousteau Society are especially important in determining the significance of this food source.

An attempt to find evidence of a "Settlement Period," to date entirely unknown, is especially important. The hypothesis that the first immigrants came as a very small group with a limited cultural inventory can only be tested here. The study recently completed by William Ayers of stratigraphic sections in a series of caves is a step in this direction. The fact that the lowest levels of these caves, which might reasonably have been expected to have attracted early immigrants, consistently produced dates significantly later than some of the developed ceremonial architecture is a significant though somewhat surprising datum suggesting that "Settlement Period" evidence is not to be sought in these contexts. The writer is inclined to interpret this evidence to mean that these caves, typically somewhat damp even under relatively dry conditions of today, were, under earlier conditions of much heavier vegetation, probably so filled with water, vegetation and dampness as not to be suitable for human habitation. This suggests that "Settlement Period" evidence is to be sought in open sites in particularly favorable shore areas such as Anakena, Ovahe, Hanga Hoonu, Hanga Nui, Akahanga and elsewhere. This poses a serious problem in that it is in precisely these areas that later ceremonial and other construction as well as other modifications of the terrain through land use have been extensive. It may be that discovery of a "Settlement Period" will come about accidently as a by-product of investigation of later architecture in these areas. Such investigations must be carried out with this facet of the problem sharply in mind. Nevertheless, direct searches for this kind of evidence must be carried out with all the ingenuity that can be brought to bear on the problem.

The testing of the hypothesis that the ceremonial architecture, from its beginning to the time of the destruction of the *ahu* and the statues, represents a continuous sequence of local development

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unmodified by sharp changes caused by outside influence needs more evidence though the nature of the picture is beginning to emerge. A recent comparative study of eleven dated *ahu* chronologically spread widely over the sequence has supported the hypothesis (Mulloy and Figueroa).

The late period of internal violence remains relatively unstudied in detail. Badly needed is a number of detailed descriptions of so-called semipyramidal *ahu*, rectangular *ahu* and *ahu poepoe* with an extensive series of dates from these. Especially important are stratigraphic studies of semipyramidal *ahu* built over earlier image *ahu*. More studies aimed at determining the exact pattern of destruction of earlier image *ahu* are needed to test the hypothesis that these were destroyed by elements of the societies of their owners rather than by enemy groups from other localities. Important here are additional detailed descriptions of the practice of deliberately breaking statues, the burying of them in *ahu* ramps, and patterns of modification apparently designed to facilitate their concealment by building semipyramidal *ahu* over them or covering them with amorphous mantles of stones.

Studies of the domestic establishments both in caves and in the open of this period are important to provide the details of their characteristics, dates of their construction, and to isolate the features that differentiate them from earlier ones. It is important to determine whether or not the elliptical, thatched houses with dressed stone foundations *(hare paenga)* of the earlier period continued to be made or whether, as seems likely, only those already present continued in use. The chronological position of stone chicken houses *(hare moa)* and protective agricultural enclosures (manavai) must be more precisely determined. Were these a distinctive characteristic of this period or were they also in use in earlier times?

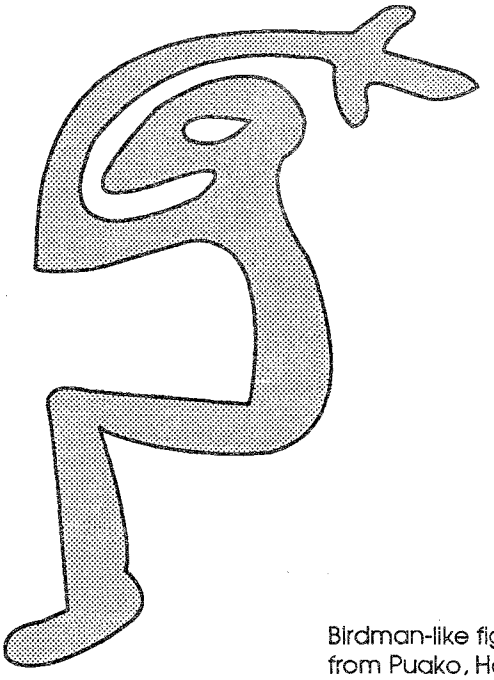
The hypothesis that there was a change in burial customs beginning with this period from cremation to inhumation requires further testing. It has often been remarked that analogy with other Polynesian practice suggests the possibility that the large amounts of burned human remains found in the crematoria behind the image *ahu* represent, not the characteristic mortuary practice of the image *ahu* period but the remains of human sacrifice or cannibalistic meals. The best way to validate this notion would be to find evidence of just what the alternative mortuary practice of the earlier period was. To date the writer knows of no conclusive evidence of tomb building or other inhumation in image *ahu* that can be demonstrated to date before the toppling of the statues and the destruction of the architecture. Also, no evidence has been found of any alternative mortuary practice other than the *ahu* crematoria which can be demonstrated to date from the time of image *ahu* building. Further evidence on these points must be carefully sought.

Much greater attention needs to be given soon to the development of a collection of dated and context-established skeletal material for physical anthropological studies. The present rapidity with which tombs are being opened and bones removed or scattered by islanders and by tourists makes this an important priority. If measures are not taken soon, most of this evidence will be lost. A permanently stationed physical anthropologist is badly needed. Such an individual could devote part of his time to collecting the easily available material, recording its context, and preparing it for study in a central repository.

To further test the hypothesis of essential cultural continuity, a much larger collection of precisely dated and context established portable artifacts must be made available. Some of this evidence is now present but much more is needed. The past emphasis on investigation and restoration of ceremonial architecture, where artifacts are usually found in secondarily deposited fill, has retarded the development of this body of evidence. More artifacts need to be collected; particularly from dated caves and open domestic establishment contexts and from stratified middens.

The above are only a few examples of hypotheses that may be formulated and tested.. There are many others. Studies aimed at testing this framework will surely modify the model and may eventually demonstrate that it should be discarded.

HAWAIIAN Rock Art Project



Birdman-like figure from Puako, Hawaii

As a complement to her six year study of the petroglyphs of Rapa Nui, Georgia Lee will begin a comparative investigation of Hawaiian petroglyphs in August-September 1988. This field season will concentrate on the island of Lana'i, which has numerous examples of figures with bird attributes. It is of interest to note that the Big Island of Hawaii has a few petroglyphs that are very close in style to the famous birdmen of Easter Island, and a boulder from Oahu, now in the Bishop Museum of Honolulu, has two bas relief figures that are astonishing close to the birdman motif. This is not to suggest that there was prehistoric contact, but is more likely an outgrowth of common ancestral backgrounds and shared myths, cosmology and religious concepts.

For information on how to participate in the project, contact University Research Expeditions Program (UREP), University of California, Berkeley, California 94720.